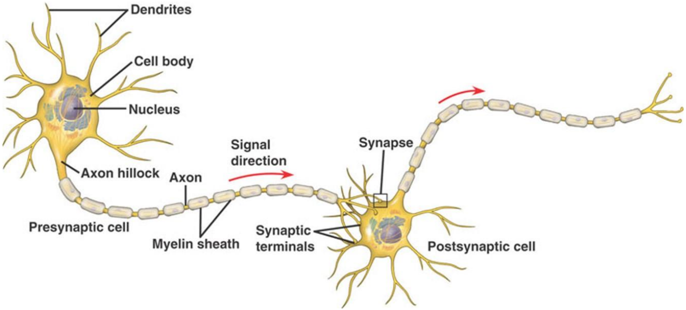
Foundations of 4D/6D Object Capture for Virtual Environments

Neural Radiance Fields

March 9, 2026

Last Lecture

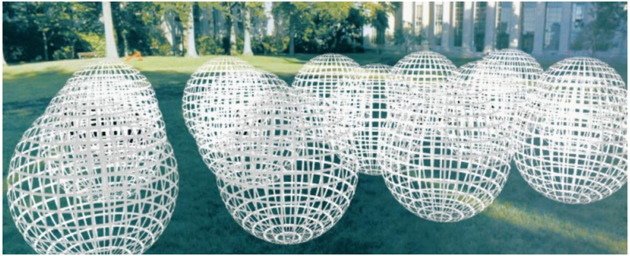
$\rightarrow$ **Deep Learning**



Today: Neural Radiance Fields

Reminder: Volumetric Videos

Question: How can we mathematically characterize the difference between 2D Videos and Volumetric Videos?



A **Volumetric Video** $P(\theta, \phi, \lambda, t, V_x, V_y, V_z)$ is the light observed . . .

- . . . from every view direction (θ, ϕ)
- . . . from every wavelength λ
- . . . at every time t
- . . . at every position (V_x, V_y, V_z) in space

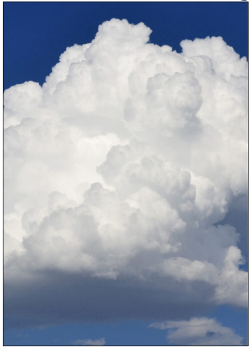
Volume Rendering

So far, we concentrated on "surface rendering", but this only covers a part of the Plenoptic Function P

Question: How can we model semi-translucent objects and effects, i.e. participating media, like clouds, fog, . . . ?

Absorption

Scattering

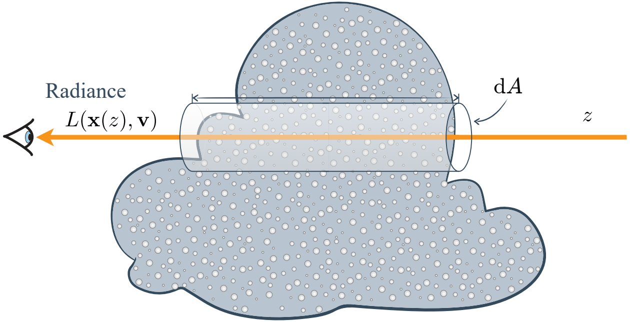


Emission

In the context of rendering, the Plenoptic Function P is typically given in terms of the radiance $\boldsymbol{L}(\boldsymbol{x}, \boldsymbol{v})$:

Scenario:

- Light travels through a differential beam with diameter $\mathrm{d}A$ in direction $\boldsymbol{v}$
- Radiance changes along z due to interactions of: photons $\leftrightarrow$ particles in participating medium
- Interactions happen *discretely*, but process can be described *stochastically*



Question: In which ways can light interact with a participating medium?

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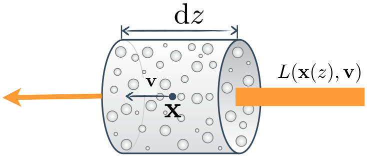
Interactions:

- Absorption
- ...



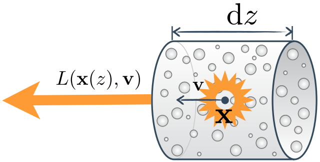
$$\frac{\mathrm{d}\boldsymbol{L}(\boldsymbol{x}(z), \boldsymbol{v})}{\mathrm{d}z} = -\mu_a(\boldsymbol{x}(z))\, \boldsymbol{L}(\boldsymbol{x}(z), \boldsymbol{v})$$

- Outgoing radiance reduced by absorption coefficient $\mu_a \geq 0$ (probability density of absorption)

Question: In which ways can light interact with a participating medium?

Interactions:

- Absorption
- Emission
- ...



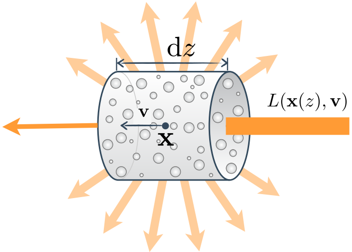
$$\frac{\mathrm{d}\boldsymbol{L}(\boldsymbol{x}(z), \boldsymbol{v})}{\mathrm{d}z} = \mu_a(\boldsymbol{x}(z))\, \boldsymbol{L}_e(\boldsymbol{x}(z), \boldsymbol{v})$$

- Outgoing radiance increased by absorption coefficient $\mu_a \geq 0$ (probability density of emission)
- Physical Motivation: Photon emission is time-reversed operation of photon absorption $\rightarrow$ same coefficient

Question: In which ways can light interact with a participating medium?

Interactions:

- Absorption
- Emission
- Out-Scattering
- ...



$$\frac{\mathrm{d}\boldsymbol{L}(\boldsymbol{x}(z), \boldsymbol{v})}{\mathrm{d}z} = -\mu_s(\boldsymbol{x}(z))\, \boldsymbol{L}(\boldsymbol{x}(z), \boldsymbol{v})$$

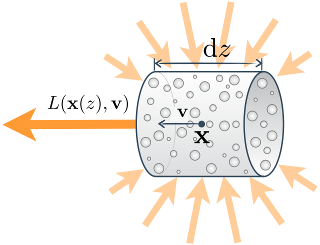
- Outgoing radiance reduced by scattering coefficient $\mu_s \geq 0$ (probability density of scattering collision)

Volume Rendering

Question: In which ways can light interact with a participating medium?

Interactions:

- Absorption
- Emission
- Out-Scattering
- *In-Scattering*

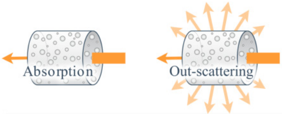


$$\frac{\mathrm{d}\boldsymbol{L}(\boldsymbol{x}(z), \boldsymbol{v})}{\mathrm{d}z} = \mu_s(\boldsymbol{x}(z))\, \boldsymbol{L}_s(\boldsymbol{x}(z), \boldsymbol{v})$$

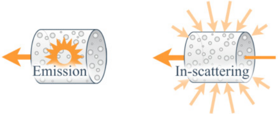
- Outgoing radiance increased by scattering coefficient $\mu_s \geq 0$ (probability density of scattering collision)
- Depends on in-scattered radiance $\boldsymbol{L}_s(\boldsymbol{x}(z), \boldsymbol{v}) = \int_{\mathbb{S}^2} \boldsymbol{f}_{\text{phase}}(\boldsymbol{x}(z), \boldsymbol{l}, \boldsymbol{v})\, \boldsymbol{L}(\boldsymbol{x}(z), \boldsymbol{l})\, \mathrm{d}\boldsymbol{l}$

Question: In which ways can light interact with a participating medium?

→ Volume Rendering can be fully described by the Radiative Transfer Equation[1]



$$\frac{\mathrm{d}L(\mathbf{x}(z),\mathbf{v})}{\mathrm{d}z} = \boxed{-\mu_a(\mathbf{x})L(\mathbf{x},\mathbf{v}) \quad -\mu_s(\mathbf{x})L(\mathbf{x},\mathbf{v}) \quad \text{Losses}} \\ \boxed{+\mu_a(\mathbf{x})L_e(\mathbf{x},\mathbf{v}) \; +\mu_s(\mathbf{x})L_s(\mathbf{x},\mathbf{v}) \quad \text{Gains}}$$



Problem: Similar to the Rendering Equation (for Surface Rendering), this formula is recursive

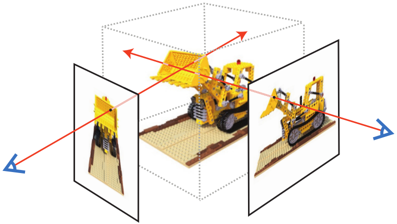
[1] S. Chandrasekhar. *Radiative transfer*. 1960.

Volume Rendering of NeRF

Problem: Similar to Rendering Equation (for Surface Rendering), this formula is recursive

Idea:

- Drop complex in-scattering and out-scattering terms
- Keep emission and absorption terms



$$\frac{\mathrm{d}\boldsymbol{L}(\boldsymbol{x}(z),\boldsymbol{v})}{\mathrm{d}z} = -\mu_a(\boldsymbol{x}(z))\,\boldsymbol{L}(\boldsymbol{x}(z),\boldsymbol{v}) + \mu_a(\boldsymbol{x}(z))\,\boldsymbol{L}_e(\boldsymbol{x}(z),\boldsymbol{v})$$

$\rightarrow$ Tracing of primary rays sufficient

This serves as the basis for what Neural Radiance Fields (NeRFs)[2] can represent!

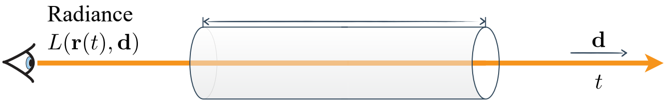
[2] B. Mildenhall et al. "NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis". In: *European Conference on Computer Vision (ECCV)*. 2020.

Volume Rendering of NeRF

In the context of NeRF, a slightly different and shorter notation is used:

$$\frac{\mathrm{d}\boldsymbol{L}(\boldsymbol{x}(z),\boldsymbol{v})}{\mathrm{d}z} = -\mu_a(\boldsymbol{x}(z))\,\boldsymbol{L}(\boldsymbol{x}(z),\boldsymbol{v}) + \mu_a(\boldsymbol{x}(z))\,\boldsymbol{L}_e(\boldsymbol{x}(z),\boldsymbol{v})$$

$$\begin{aligned} t &:= -z \\ \boldsymbol{r}(t) &:= \boldsymbol{x}(-z) \\ \boldsymbol{d} &:= \boldsymbol{v} \\ \sigma(\boldsymbol{r}(t)) &:= \mu_a(\boldsymbol{x}(-z)) \\ \hat{\boldsymbol{C}}(\boldsymbol{r}(t),\boldsymbol{d}) &:= \boldsymbol{L}(\boldsymbol{x}(-z),\boldsymbol{v}) \\ \boldsymbol{c}(\boldsymbol{r}(t),\boldsymbol{d}) &:= \boldsymbol{L}_e(\boldsymbol{x}(-z),\boldsymbol{v}) \end{aligned} \quad\Rightarrow\quad \begin{aligned} \sigma(t) &:= \sigma(\boldsymbol{r}(t)) \\ \boldsymbol{c}(t) &:= \boldsymbol{c}(\boldsymbol{r}(t),\boldsymbol{d}) \\ \hat{\boldsymbol{C}}(t) &:= \hat{\boldsymbol{C}}(\boldsymbol{r}(t),\boldsymbol{d}) \end{aligned} \quad\Rightarrow\quad \frac{\mathrm{d}\hat{\boldsymbol{C}}(t)}{\mathrm{d}t} = \sigma(t)\,\hat{\boldsymbol{C}}(t) - \sigma(t)\,\boldsymbol{c}(t)$$



Note: Sign flips because the direction $t := -z$ has changed!

Volume Rendering of NeRF

The solution of this differential equation is the following simplified version of the Volume Rendering Equation:

$$\hat{\boldsymbol{C}}(t) = \int_t^{t_{\text{end}}} \underbrace{T(t \to s)}_{\substack{\text{visibility of } \boldsymbol{r}(s) \\ \text{from } \boldsymbol{r}(t)}} \underbrace{\sigma(s)\, \boldsymbol{c}(s)}_{\substack{\text{light emission} \\ \text{from } \boldsymbol{r}(s)}} \,\mathrm{d}s + T(t \to t_{\text{end}})\, \boldsymbol{C}_{\text{const}}$$

Here, the Transmittance factor T is defined as

$$T(a \to b) = \mathrm{e}^{-\int_a^b \sigma(u)\,\mathrm{d}u}$$

Properties:

Derivatives

$$\frac{\mathrm{d}}{\mathrm{d}t} T(t \to s) = \quad \sigma(t)\, T(t \to s)$$

$$\frac{\mathrm{d}}{\mathrm{d}s} T(t \to s) = -\sigma(s)\, T(t \to s)$$

Antiderivative

$$\int_a^b \sigma(s)\, T(t \to s)\,\mathrm{d}s$$

$$= T(t \to a) - T(t \to b)$$

Multiplicative

$$T(a \to c) = T(a \to b)\, T(b \to c)$$

Volume Rendering of NeRF

Sanity Check: We should validate that this is indeed a solution for the Radiative Transfer Equation

$$\frac{\mathrm{d}\hat{\boldsymbol{C}}(t)}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\int_t^{t_{\text{end}}} T(t \to s)\,\sigma(s)\,\boldsymbol{c}(s)\,\mathrm{d}s + T(t \to t_f)\,\hat{\boldsymbol{C}}_{\mathrm{const}}\right)$$

$$= \underbrace{\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_t^{t_{\text{end}}} T(t \to s)\,\sigma(s)\,\boldsymbol{c}(s)\,\mathrm{d}s\right)}_{\text{foreground term}} + \underbrace{\frac{\mathrm{d}}{\mathrm{d}t}\left(T(t \to t_{\text{end}})\,\hat{\boldsymbol{C}}_{\mathrm{const}}\right)}_{\text{background term}}$$

1. Consider the background term:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(T(t \to t_{\text{end}})\,\hat{\boldsymbol{C}}_{\mathrm{const}}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(T(t \to t_{\text{end}})\right)\hat{\boldsymbol{C}}_{\mathrm{const}}$$

$$= \sigma(t)\,T(t \to t_{\text{end}})\,\hat{\boldsymbol{C}}_{\mathrm{const}}$$

[Derivative Property of Transmittance]

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2. Consider the foreground term:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_t^{t_{\text{end}}} T(t \to s)\,\sigma(s)\,\boldsymbol{c}(s)\,\mathrm{d}s\right)$$

$$= -\underbrace{T(t \to t)}_{=1}\,\sigma(t)\,\boldsymbol{c}(t) + \int_t^{t_{\text{end}}} \frac{\mathrm{d}}{\mathrm{d}t}\Big(T(t \to s)\,\sigma(s)\,\boldsymbol{c}(s)\Big)\,\mathrm{d}s$$

$$\left[\begin{array}{c}\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_x^b f(x,t)\,\mathrm{d}t\right) \\ = -f(x,x) + \int_x^b \frac{\mathrm{d}}{\mathrm{d}x} f(x,t)\,\mathrm{d}t\end{array}\right]$$

$$= -\sigma(t)\,\boldsymbol{c}(t) + \int_t^{t_{\text{end}}} \frac{\mathrm{d}}{\mathrm{d}t}\Big(T(t \to s)\Big)\,\sigma(s)\,\boldsymbol{c}(s)\,\mathrm{d}s$$

$$= -\sigma(t)\,\boldsymbol{c}(t) + \int_t^{t_{\text{end}}} \sigma(t)\,T(t \to s)\,\sigma(s)\,\boldsymbol{c}(s)\,\mathrm{d}s$$

[Derivative Property of Transmittance]

$$= -\sigma(t)\,\boldsymbol{c}(t) + \sigma(t)\int_t^{t_{\text{end}}} T(t \to s)\,\sigma(s)\,\boldsymbol{c}(s)\,\mathrm{d}s$$

$$= -\sigma(t)\,\boldsymbol{c}(t) + \sigma(t)\left[\hat{\boldsymbol{C}}(t) - T(t \to t_{\text{end}})\,\hat{\boldsymbol{C}}_{\text{const}}\right]$$

[Definition of Volume Rendering Equation]

Volume Rendering of NeRF

3. Combine both terms two together:

$$\frac{\mathrm{d}\hat{\boldsymbol{C}}(t)}{\mathrm{d}t} = \underbrace{\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_t^{t_{\mathsf{end}}} T(t \to s)\,\sigma(s)\,\boldsymbol{c}(s)\,\mathrm{d}s\right)}_{\text{foreground term}} + \underbrace{\frac{\mathrm{d}}{\mathrm{d}t}\left(T(t \to t_{\mathsf{end}})\,\hat{\boldsymbol{C}}_{\mathrm{const}}\right)}_{\text{background term}}$$

$$= -\sigma(t)\,\boldsymbol{c}(t) + \sigma(t)\left[\hat{\boldsymbol{C}}(t) - T(t \to t_{\mathsf{end}})\,\hat{\boldsymbol{C}}_{\mathrm{const}}\right] + \sigma(t)\,T(t \to t_{\mathsf{end}})\,\hat{\boldsymbol{C}}_{\mathrm{const}}$$

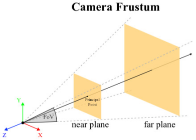
$$= -\sigma(t)\,\boldsymbol{c}(t) + \sigma(t)\,\hat{\boldsymbol{C}}(t)$$

$\rightarrow$ This is exactly the Radiative Transfer Equation ✓

For practical rendering, NeRF's volume rendering formulation needs a few more adjustments:

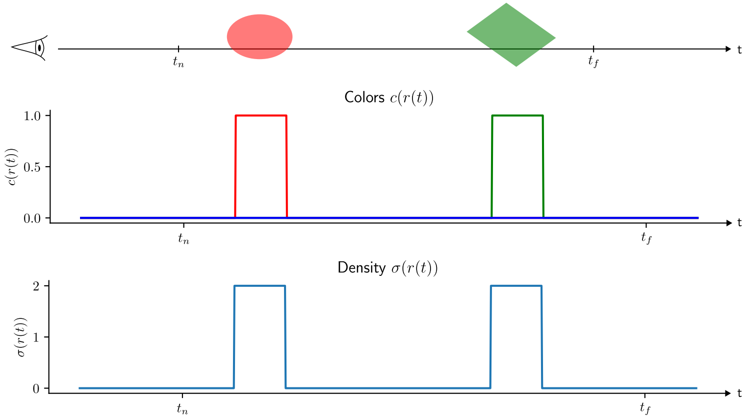
- Want color at near plane $t = t_n$ (image plane), and clip at far plane $t_{\mathsf{end}} = t_f$
- Use background color $\hat{\boldsymbol{C}}_{\mathrm{bg}}$

$$\hat{\boldsymbol{C}} = \int_{t_n}^{t_f} T(t_n \to s)\,\sigma(s)\,\boldsymbol{c}(s)\,\mathrm{d}s + T(t_n \to t_f)\,\hat{\boldsymbol{C}}_{\mathrm{bg}}$$



Volume Rendering of NeRF

Example: Two objects with different colors but equal density



What will be the *dominant color tone* of the final color $\hat{C}$ blended along the ray?

Example: Two objects with different colors but equal density

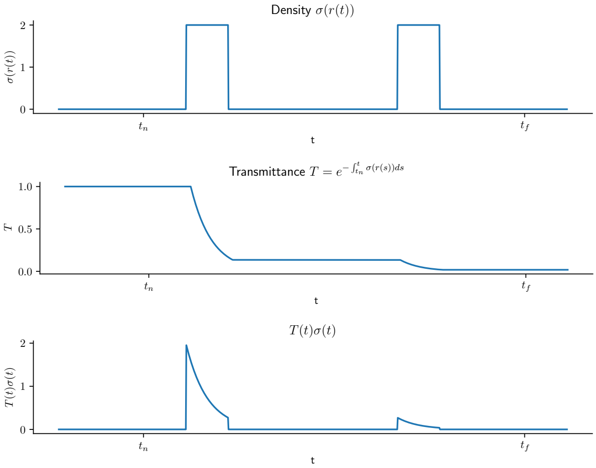
Transmittance:

- Starts at 1
- Monotonically decreases
- Reaches small "remainder" value ϵ

Blending weights: $T(t)\,\sigma(t)$

- Same "hill locations" as density
- Less influence of object "backside" than front part

$\rightarrow$ As expected, the final color $\hat{C}$ will be mostly red



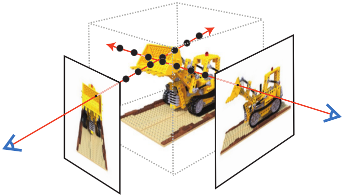
Volume Rendering of NeRF

Problem: Integrals over σ, T and $\boldsymbol{c}$ cannot be calculated analytically

$$\hat{\boldsymbol{C}} = \int_{t_n}^{t_f} T(t_n \to s)\,\sigma(s)\,\boldsymbol{c}(s)\,\mathrm{d}s + T(t_n \to t_f)\,\hat{\boldsymbol{C}}_{\mathrm{bg}}$$

Solution: Assume that density $\sigma(t)$ and color $\boldsymbol{c}(t)$ are piecewise constant:

$$\sigma(t) = \sigma_i \quad \forall t \in [t_i, t_{i+1}) \qquad , \qquad \boldsymbol{c}(t) = \boldsymbol{c}_i \quad \forall t \in [t_i, t_{i+1})$$



- Divide ray into N evenly sized bins
- Uniformly sample one point from each bin $\rightarrow t_n \leq t_1 < \cdots < t_N \leq t_f$ with distances $\delta_i = t_{i+1} - t_i$

Volume Rendering of NeRF

With this assumption, solve the integral of the foreground term:

$$\int_{t_n}^{t_f} T(t_n \to s)\,\sigma(s)\,\boldsymbol{c}(s)\,\mathrm{d}s = \sum_{i=1}^{N} \int_{t_i}^{t_{i+1}} T(t_1 \to s)\,\sigma(s)\,\boldsymbol{c}(s)\,\mathrm{d}s \qquad \text{[Split integral]}$$

$$= \sum_{i=1}^{N} \left(\int_{t_i}^{t_{i+1}} T(t_1 \to s)\,\sigma(s)\,\mathrm{d}s \right) \boldsymbol{c}_i \qquad \text{[Piecewise constant color]}$$

$$= \sum_{i=1}^{N} \big(T(t_1 \to t_i) - T(t_1 \to t_{i+1}) \big)\,\boldsymbol{c}_i \qquad \text{[Antiderivative Property of Transmittance]}$$

$$= \sum_{i=1}^{N} \big(T(t_1 \to t_i) - T(t_1 \to t_i)\,T(t_i \to t_{i+1}) \big)\,\boldsymbol{c}_i \qquad \text{[Multiplicative Property]}$$

$$= \sum_{i=1}^{N} T_i \left(1 - \mathrm{e}^{-\sigma_i \delta_i}\right) \boldsymbol{c}_i \qquad \text{[Piecewise constant density]}$$

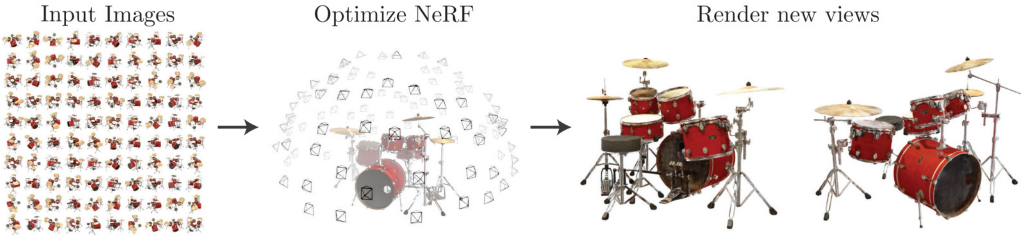
Volume Rendering of NeRF

Thus, NeRF-style Volumetric Rendering is performed by computing

$$\hat{C} = \sum_{i=1}^{N} T_i \left(1 - \mathrm{e}^{-\sigma_i \delta_i}\right) \boldsymbol{c}_i + T_{N+1}\, \hat{C}_{\mathrm{bg}} \qquad , \qquad T_i = \mathrm{e}^{-\sum_{j=1}^{i-1} \sigma_j\, \delta_j}$$

Idea:

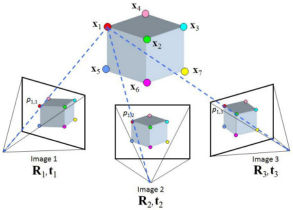
- **Neural:** Use Deep Learning approach as underlying model
- **Radiance:** Estimate radiance of scene
- **Field:** Represent scene volumetrically

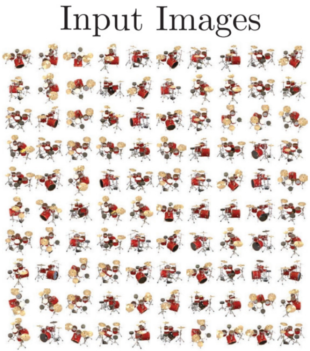


NeRF: Input

As input, NeRF expects a set of images from various positions and angles:

- RGB images C_k
- Camera Intrinsics K
- Camera Extrinsics T_k



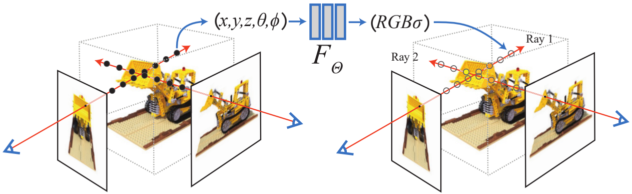


"Standard Method": Camera parameters estimated via Structure from Motion (SfM) $\rightarrow$ COLMAP[3]

[3] J. L. Schönberger and J.-M. Frahm. "Structure-from-Motion Revisited". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2016.

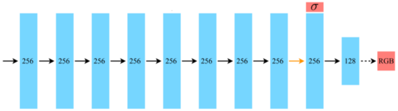
NeRF: Scene Representation

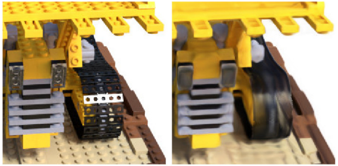
Question: How can we represent the scene as a "radiance field"?



Use a large MLP F_Θ

- Input: Position $\boldsymbol{x}$ and direction $\boldsymbol{d}$
- Output: Radiance $\boldsymbol{c}$ and density σ





Ground Truth

Direct Input

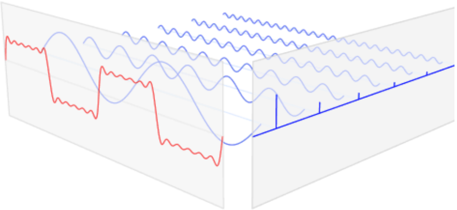
→ High-frequency color details from direct inputs $\boldsymbol{x}$ and $\boldsymbol{d}$ not captured and smoothed away

NeRF: Positional Encoding

In order to retain high-frequency details, we need to consider the Fourier Transform:

Spatial domain

$$f(t) = \int_{\mathbb{R}} \widehat{f}(\omega)\, \mathrm{e}^{2\pi \mathrm{i} \omega t}\, \mathrm{d}\omega$$



Frequency domain

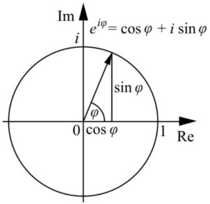
$$\widehat{f}(\omega) = \int_{\mathbb{R}} f(t)\, \mathrm{e}^{-2\pi \mathrm{i} \omega t}\, \mathrm{d}t$$

Lemma (Euler's formula)

The following equation is called Euler's formula:

$$\mathrm{e}^{\mathrm{i}x} = \cos(x) + \mathrm{i}\sin(x)$$

$\rightarrow$ Fourier Transform decomposes f to sine and cosine functions



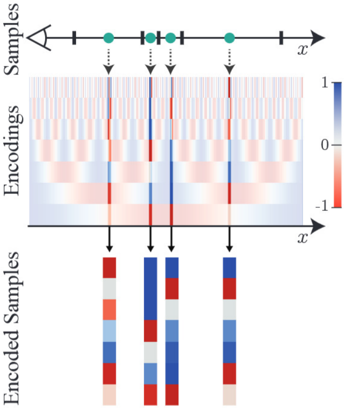
NeRF: Positional Encoding

Idea: Provide frequency information to MLP $\rightarrow$ Positional Encoding

$$\gamma(\boldsymbol{p}) = \left(\sin\left(2^0 \pi\, \boldsymbol{p}\right), \cos\left(2^0 \pi\, \boldsymbol{p}\right), \ldots, \sin\left(2^{L-1} \pi\, \boldsymbol{p}\right), \cos\left(2^{L-1} \pi\, \boldsymbol{p}\right)\right)^T \in \mathbb{R}^{2L \cdot 3}$$

- Position $\boldsymbol{x} \in \mathbb{R}^3$: $L = 10$ frequencies
- Direction $\boldsymbol{d} \in \mathbb{R}^3$: $L = 4$ frequencies

$\rightarrow$ Small coordinate differences between two points are spread and amplified in different frequency bands

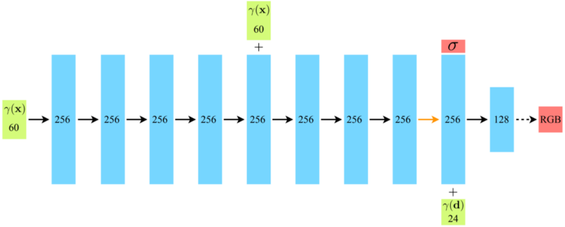


NeRF: Positional Encoding

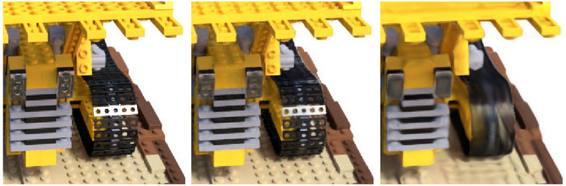
Idea: Provide frequency information to MLP → Positional Encoding

MLP Architecture:

- Fully connected layers
- ReLU activations
- No/Linear activation at 8th layer (orange arrow)
- Encoded position $\gamma(\boldsymbol{x})$ and direction $\gamma(\boldsymbol{d})$ concatenated ($+$ sign)

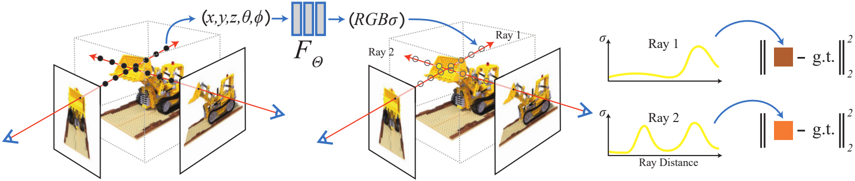


→ High-frequency color details can be learned with positional encoding


Ground Truth Complete Model No Positional Encoding

NeRF: Training

Now, we can fit all the pieces together for training a NeRF



Forward Pass:

1. March camera rays from inputs $\boldsymbol{K}$, $\boldsymbol{T}_k$
2. (Uniformly) sample points $\boldsymbol{x}$ along the ray with direction $\boldsymbol{d}$
3. Evaluate the MLP at the samples to get values for radiance $\boldsymbol{c}$ and density σ
4. Accumulate values to obtain volume rendered images $\hat{\boldsymbol{C}}_k$

Backward Pass: Minimize image loss to input images $\boldsymbol{C}_k$ $\rightarrow$ Entire pipeline fully differentiable

$$\mathcal{L}(\boldsymbol{\Theta}) = \sum_k \left\| \boldsymbol{C}_k - \hat{\boldsymbol{C}}_k(\boldsymbol{\Theta}) \right\|^2$$

NeRF: Results

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Even complex scene representations can be learnt

Interactive Results: `https://www.matthewtancik.com/nerf`

NeRF: Results

Even complex scene representations can be learnt

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NeRF: Results

Advantages

- Very detailed results and realistic novel view synthesis
- Glossy surfaces can also be modelled
- 3D geometry/Depth can be estimated from density σ

$\rightarrow$ NeRF can represent a **Volumetric Image**

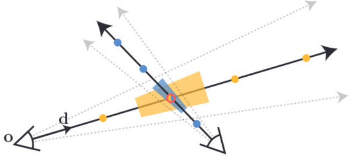
$$P(\theta, \phi, \lambda, V_x, V_y, V_z)$$

Limitations

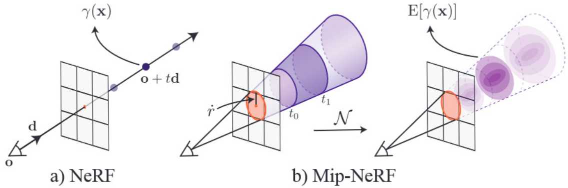
- Aliasing artifacts at fine structures
  $\rightarrow$ Mip-NeRF
- Training time with $\sim$ 12 hours per scene
  $\rightarrow$ Instant NGP
- Rendering time with $\sim$ 30 seconds per frame
  $\rightarrow$ Gaussian Splatting
- Static scenes only
  $\rightarrow$ D-NeRF

Extensions: Mip-NeRF

Problem: Traversing single ray per pixel ignore size and shape of sampled volume → blurred and aliased images[4]



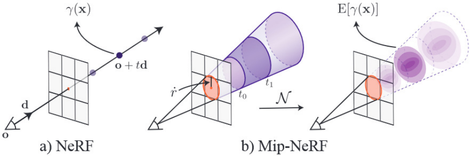
Idea: Render conical frustums while marching the ray

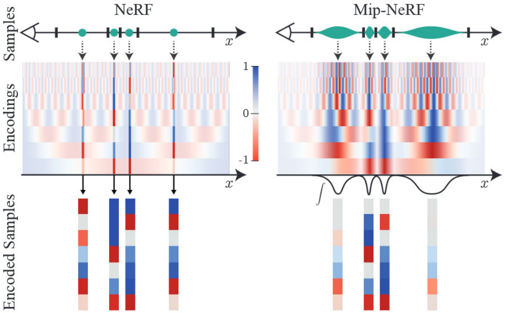


[4] J. T. Barron et al. "Mip-NeRF: A Multiscale Representation for Anti-Aliasing Neural Radiance Fields". In: *IEEE/CVF International Conference on Computer Vision (CVPR)*. 2021.

Extensions: Mip-NeRF

Idea: Render conical frustums while marching the ray





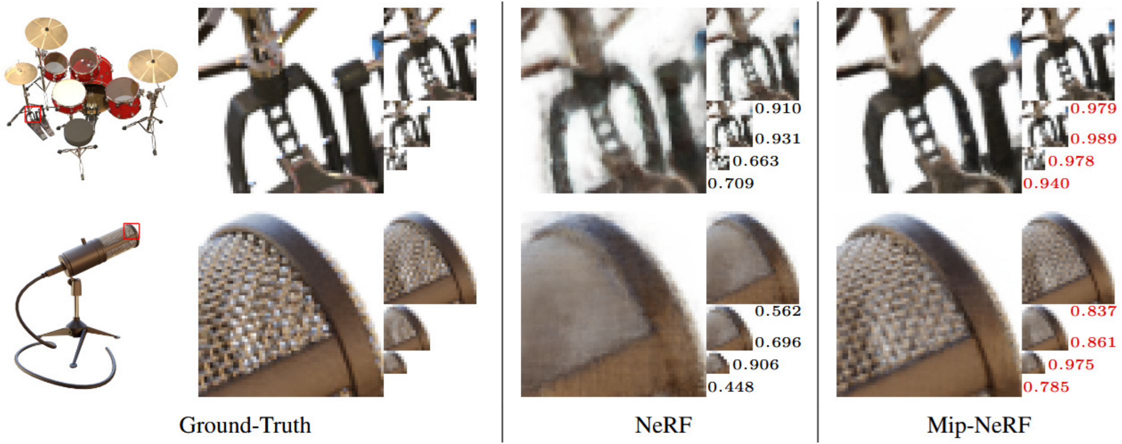
- Model each cone segment by normal distribution $\mathcal{N}(\cdot \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$
- Only positional encoding needs to be adapted

$\rightarrow$ Low-pass filter

$$\gamma(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \mathbb{E}[\gamma(\boldsymbol{x})]$$

$$= \left(\dots, \sin\left(2^l \boldsymbol{\mu}\right) \cdot \mathrm{e}^{-\frac{1}{2} 4^l \operatorname{diag}(\boldsymbol{\Sigma})}, \cos\left(2^l \boldsymbol{\mu}\right) \cdot \mathrm{e}^{-\frac{1}{2} 4^l \operatorname{diag}(\boldsymbol{\Sigma})}, \dots \right)^T$$

Results:



Interactive Results: `https://jonbarron.info/mipnerf`

Extensions: Instant NGP

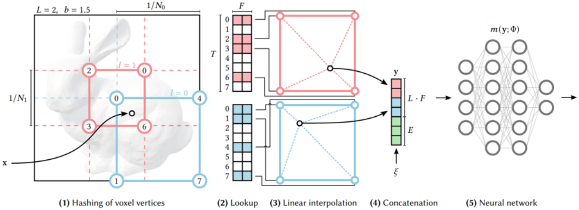
Problem: Very long training times → Size of MLP with millions of learnable parameters[5]

Idea: Hash Encoding

1. Divide space into multi-resolution voxel grid
2. Find voxel of $\boldsymbol{p}$ in each level
3. Lookup <u>learnable feature vectors</u> $\boldsymbol{F}$ (with trilinear interpolation) from hash table

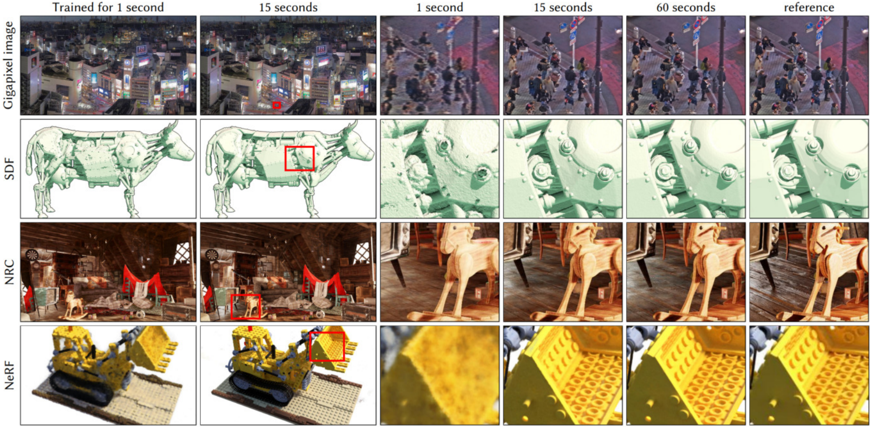
   $$H(x, y, z) = (x \cdot p_1 \oplus y \cdot p_2 \oplus z \cdot p_3) \bmod n$$

   → Same scheme as in VoxelHashing
4. Evaluate concatenated features by <u>small</u> MLP → density σ and radiance $\boldsymbol{c}$



[5] T. Müller et al. "Instant Neural Graphics Primitives with a Multiresolution Hash Encoding". In: *ACM Transactions on Graphics (TOG)* 41.4 (2022).

Results:



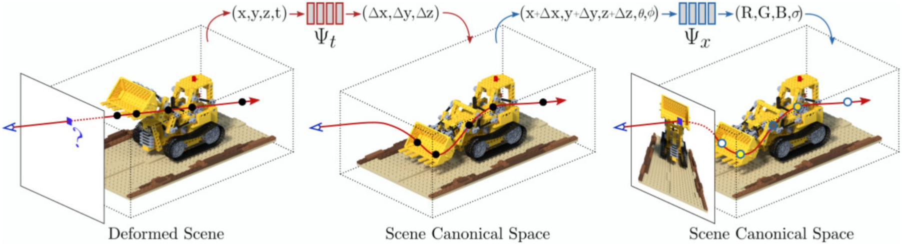
Interactive Results: `https://nvlabs.github.io/instant-ngp`

Extensions: D-NeRF

Problem: NeRF can only handle static scenes $\rightarrow$ 1 NeRF per frame in naive way[6]

Idea:

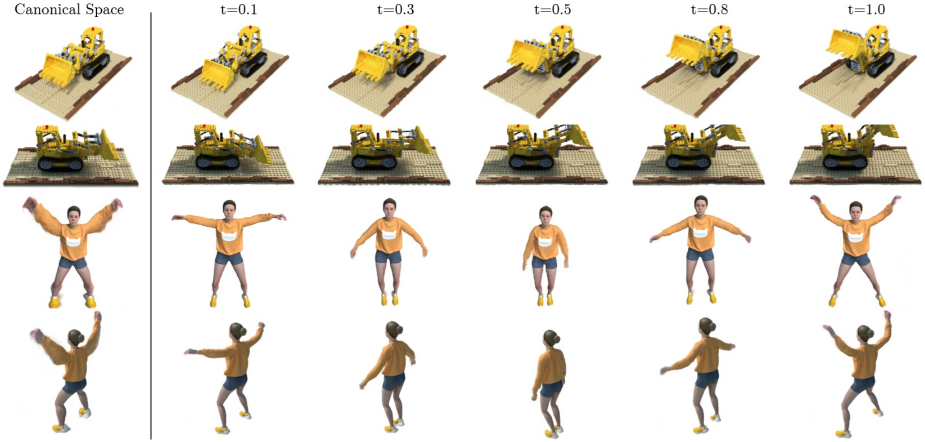
- Train 2 networks: 1 for the deformation + 1 for the reference static NeRF ("canonical NeRF")
- Warp sampled points x by deformation network to get time-dependent offset Δx
- Query canonical NeRF and perform volume rendering as usual



[6] A. Pumarola et al. “D-NeRF: Neural Radiance Fields for Dynamic Scenes”. In: *IEEE/CVF International Conference on Computer Vision (CVPR)*. 2021.

Extensions: D-NeRF

Results:

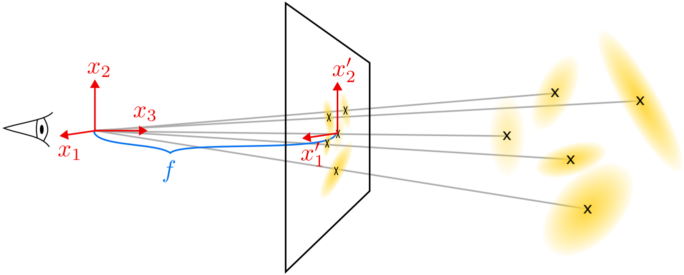


Interactive Results: `https://www.albertpumarola.com/research/D-NeRF/index.html`

Next Lecture: Gaussian Splatting

Next Lecture

→ **Gaussian Splatting**



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[1] J. T. Barron, B. Mildenhall, M. Tancik, P. Hedman, R. Martin-Brualla, and P. P. Srinivasan. "Mip-NeRF: A Multiscale Representation for Anti-Aliasing Neural Radiance Fields". In: *IEEE/CVF International Conference on Computer Vision (CVPR)*. 2021.

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[4] T. Müller, A. Evans, C. Schied, and A. Keller. "Instant Neural Graphics Primitives with a Multiresolution Hash Encoding". In: *ACM Transactions on Graphics (TOG)* 41.4 (2022).

[5] A. Pumarola, E. Corona, G. Pons-Moll, and F. Moreno-Noguer. "D-NeRF: Neural Radiance Fields for Dynamic Scenes". In: *IEEE/CVF International Conference on Computer Vision (CVPR)*. 2021.

[6] J. L. Schönberger and J.-M. Frahm. "Structure-from-Motion Revisited". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2016.